Efficient Multi-Label Classification of 12-Lead ECG Signals Using Deep Learning with Pruning and Quantization

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Abstract

Automated classification of 12-lead electrocardiogram (ECG) signals can support efficient cardiovascular disease diagnosis. However, deep learning models may require substantial computational and memory resources. This study investigates efficient multi-label ECG classification using a one-dimensional Convolutional Neural Network (CNN) combined with model improvement and compression techniques. Binary Cross-Entropy (BCE), Weighted BCE, Focal Loss, Squeeze-and-Excitation (SE) attention, data augmentation, pruning, and dynamic INT8 quantization were evaluated. The CNN trained with BCE achieved the highest validation accuracy of 95.22%, while Weighted BCE achieved the highest validation AUROC of 0.8775. On the independent test set, the selected FP32 model achieved 91.42% accuracy and 0.8688 macro AUROC. Dynamic INT8 quantization achieved 91.44% accuracy and 0.8690 AUROC while reducing model size from 12.66 MB to 12.23 MB.

1 Introduction

Cardiovascular diseases remain a major global health challenge, and the electrocardiogram (ECG) is widely used for detecting cardiac abnormalities. Recent advances in deep learning have demonstrated strong performance in automated ECG classification. However, deep learning models can contain millions of parameters and require substantial computational and memory resources, which can limit deployment in resource-constrained healthcare environments.

This study investigates multi-label classification of 12-lead ECG signals using a one-dimensional CNN and evaluates several strategies for improving and compressing the model. The study specifically examines different loss functions, Squeeze-and-Excitation (SE) attention, data augmentation, unstructured pruning, and dynamic INT8 quantization. The objective is to identify a model that maintains strong classification performance while reducing computational and storage requirements.

2 Method

Data. The study used 12-lead ECG recordings sampled at 500 Hz with approximately 10 seconds of signal data per record. Each ECG was therefore represented as a multivariate time-series signal rather than an image. The signals were processed directly using a one-dimensional CNN.

Model. A one-dimensional CNN was developed for multi-label diagnosis prediction from the 12 ECG leads. The model learns temporal and morphological patterns directly from the raw ECG signals. Binary Cross-Entropy (BCE) was initially used as the main training objective.

Loss Functions and Model Improvements. To investigate the effect of class imbalance, three loss configurations were evaluated: standard BCE, Weighted BCE, and Focal Loss. Squeeze-and-

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Excitation (SE) attention and ECG data augmentation were subsequently evaluated as additional model improvements. Class-specific decision thresholds were optimized using the validation set.

Compression. Model efficiency was investigated using independent unstructured pruning experiments at 30%, 50%, and 70% sparsity followed by fine-tuning. Dynamic INT8 quantization was also applied. Finally, a combined 30% pruning and INT8 quantization experiment was evaluated.

3 Results and Findings

Table 1 summarizes the model development and compression experiments.

Table 1: Comparison of model development and compression experiments.

Stage	Model	Accuracy	AUROC	Size
Loss	CNN + BCE	**95.22%**	0.8757	–
Loss	CNN + Weighted BCE	85.38%	**0.8775**	–
Loss	CNN + Focal	94.57%	0.8670	–
Improvement	+ SE Attention	95.13%	–	–
Improvement	+ SE + Augmentation	95.20%	–	–
Test	Best FP32 CNN	91.42%	0.8688	12.66 MB
Pruning	30% Pruned	89.88%	0.8663	12.66 MB
Pruning	50% Pruned	85.77%	0.8670	12.66 MB
Pruning	70% Pruned	82.66%	0.8616	12.66 MB
Quantization	INT8	**91.44%**	**0.8690**	**12.23 MB**
Quantization	30% Pruned + INT8	89.87%	0.8662	12.23 MB

During model selection, the baseline CNN trained with BCE achieved the highest validation accuracy of 95.22%, while Weighted BCE achieved the highest AUROC of 0.8775. However, Weighted BCE substantially reduced accuracy to 85.38%. Focal Loss achieved 94.57% accuracy and an AUROC of 0.8670.

Adding SE attention and data augmentation did not improve validation accuracy compared with the BCE configuration. SE attention achieved 95.13%, while SE combined with augmentation achieved 95.20%. Therefore, the plain CNN with BCE was selected as the best configuration for independent test evaluation.

On the independent test set, the selected FP32 CNN achieved 91.42% accuracy and a macro AUROC of 0.8688. For pruning, 30% sparsity provided the best performance trade-off, achieving 89.88% accuracy and 0.8663 AUROC. Increasing sparsity to 50% and 70% resulted in progressively greater performance degradation.

Dynamic INT8 quantization achieved the best overall test performance, with 91.44% accuracy and 0.8690 AUROC, while reducing model size from 12.66 MB to 12.23 MB. The combined 30% pruning and INT8 quantization model achieved 89.87% accuracy and 0.8662 AUROC.

Pruning alone did not reduce the stored model size because unstructured pruning retained dense tensors containing zero-valued weights. Consequently, although the number of non-zero parameters decreased, the dense model representation remained approximately the same size. In contrast, INT8 quantization reduced the numerical precision of model parameters and therefore provided a measurable reduction in storage size.

4 Conclusion

The proposed one-dimensional CNN achieved strong performance for multi-label classification of 12-lead ECG signals processed directly as time-series data. Among the evaluated training configurations, BCE provided the highest validation accuracy, while Weighted BCE produced the highest validation

AUROC. However, the lower accuracy obtained with Weighted BCE led to the selection of the standard BCE configuration.

Moderate pruning demonstrated that model sparsity can be introduced while preserving relatively strong predictive performance, although aggressive pruning caused substantial performance degradation. Dynamic INT8 quantization provided the most favorable compression result, maintaining classification performance while reducing model size.

These findings indicate that quantization is a promising approach for deploying deep learning-based ECG classification systems in resource-constrained healthcare environments. Future work will investigate convolution-aware quantization, structured pruning, quantization-aware training, and sparse inference to achieve greater compression and computational efficiency.

References

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